227268: altered volcaniclastic rock, Woolgah Hill

(Polelle Group, Murchison Supergroup, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Yalgoo (SH 50-2), Yalgoo (2241)
MGA Zone 50, 469409E 6881257N

Sampled on 10 May 2019

This sample was collected from an outcrop on Gabyon Station, about 8.0 km east of Yilgiddie No. 1 Well, 6.3 km south-southeast of Warbadoo Hill, and 5.9 km northwest of Woolgah Hill.

Geological context

The unit sampled is a volcaniclastic rock of the Polelle Group of the Murchison Supergroup (Ivanic et al., 2015). The 2800–2735 Ma Polelle Group consists of metamorphosed basalt, komatiitic basalt, felsic volcanic and volcaniclastic rocks, and banded iron-formation (Van Kranendonk et al., 2013). The volcaniclastic unit strikes north-northeast near the northwestern margin of the Yalgoo–Singleton greenstone belt, and includes meta-andesite to basaltic meta-andesite with minor metabasalt and metadacite (Ivanic et al., 2015). This unit is interpreted to stratigraphically overlie basaltic rocks of the Polelle Group to the east, including a volcaniclastic metasandstone of the Yalgoo Formation, about 4.2 km to the southeast, from which detrital zircons provided a maximum depositional age of 2793 ± 8 Ma (GSWA 207686, Lu et al., 2016). A felsic metavolcaniclastic rock, sampled about 43 km to the south-southwest, yielded an igneous crystallization age of 2941 ± 3 Ma (GSWA 207670, Wingate et al., 2019).

Petrographic description

In outcrop, the rock is a strongly altered volcaniclastic conglomerate (Fig. 1), consisting of cm-scale ovoid dacite clasts in a sand–silt matrix. Microscopic examination indicates the sample is very highly altered, diffusely layered, and consists of about 50–55% quartz, 30–35% epidote, 5–10% actinolite, <5% plagioclase, and <1% titanite. There is also a small amount (~5% of the slide) of fine-grained amphibolite at one end of the thin section. Quartz forms an intergrown mosaic of anhedral, weakly strained grains up to 0.2 mm in size. The presence of rare twinned grains indicates a small amount of plagioclase. Epidote is anhedral, colourless, and forms granular masses that are disseminated throughout the quartz-rich mosaic and also in more intense narrow bands that define a discontinuous layering. Some epidote occurs in aggregates and veins of anhedral grains up to 1 mm in size. These aggregates also contain most of the pale green actinolite, which forms very fine-grained fibrous masses up to 1 mm across. Fibrous aggregates of actinolite also occur mainly in the more intensely epidote-altered layers. Crystals (<0.1 mm) and aggregates of titanite also occur where epidote alteration is most intense. The amphibolite portion of the sample consists of a microcrystalline polygonal aggregate of subequal amounts of microcrystalline to fine-grained (up to 0.5 mm) quartz and feldspar, plus 20% well-formed, blue–green to olive–green, subhedral hornblende laths up to 1 mm long, and minor titanite, all overprinted by patches of secondary epidote. The coarser grained parts include finely micropoikilitic probable plagioclase, rare twinned plagioclase and some microperthitic K-feldspar.

Zircon morphology

Only six zircons were isolated from this sample, and they are colourless to dark brown and anhedral to euhedral. The crystals are up to 100 µm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and one has a narrow luminescent rim. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed on 10 September 2020, using SHRIMP-B. Six analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.31% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.23%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Six analyses were obtained from 6 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Figure 1. **Hand specimen image for sample 227268: altered volcaniclastic rock, Woolgah Hill**

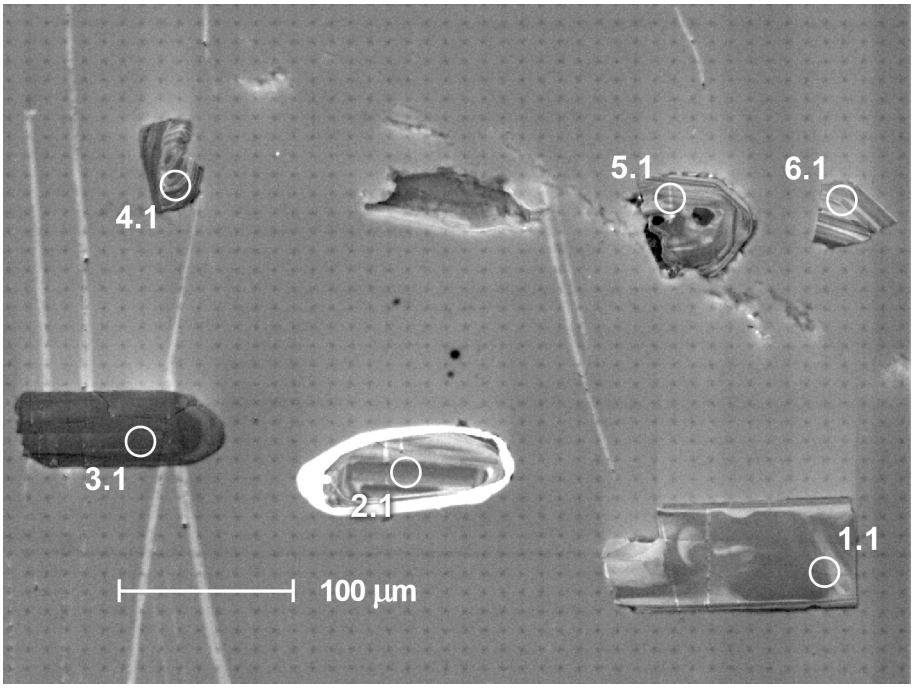

Figure 2. **Cathodoluminescence image of all zircons from sample 227268: altered volcaniclastic rock, Woolgah Hill. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 227268: altered volcaniclastic rock, Woolgah Hill

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
Y	4	4.1	443	226	0.51	0.037	1.756	0.016	0.21459	0.00091	1.756	0.016	0.21427	0.00092	2905	22	2938	7	1.1
S	6	6.1	127	72	0.56	0.000	1.699	0.022	0.21620	0.00137	1.699	0.022	0.21620	0.00137	2984	31	2953	10	-1.1
S	3	3.1	382	220	0.58	0.035	1.768	0.016	0.21662	0.00094	1.768	0.016	0.21631	0.00095	2890	22	2953	7	2.2
S	5	5.1	117	75	0.64	0.074	1.817	0.024	0.21711	0.00145	1.819	0.024	0.21646	0.00149	2825	31	2955	11	4.4
Z	2	2.1	217	141	0.65	0.050	3.462	0.037	0.10265	0.00103	3.463	0.037	0.10222	0.00107	1635	16	1665	19	1.8
D	1	1.1	173	145	0.84	0.564	2.777	0.032	0.17815	0.00243	2.792	0.032	0.17311	0.00264	1973	20	2588	25	23.8

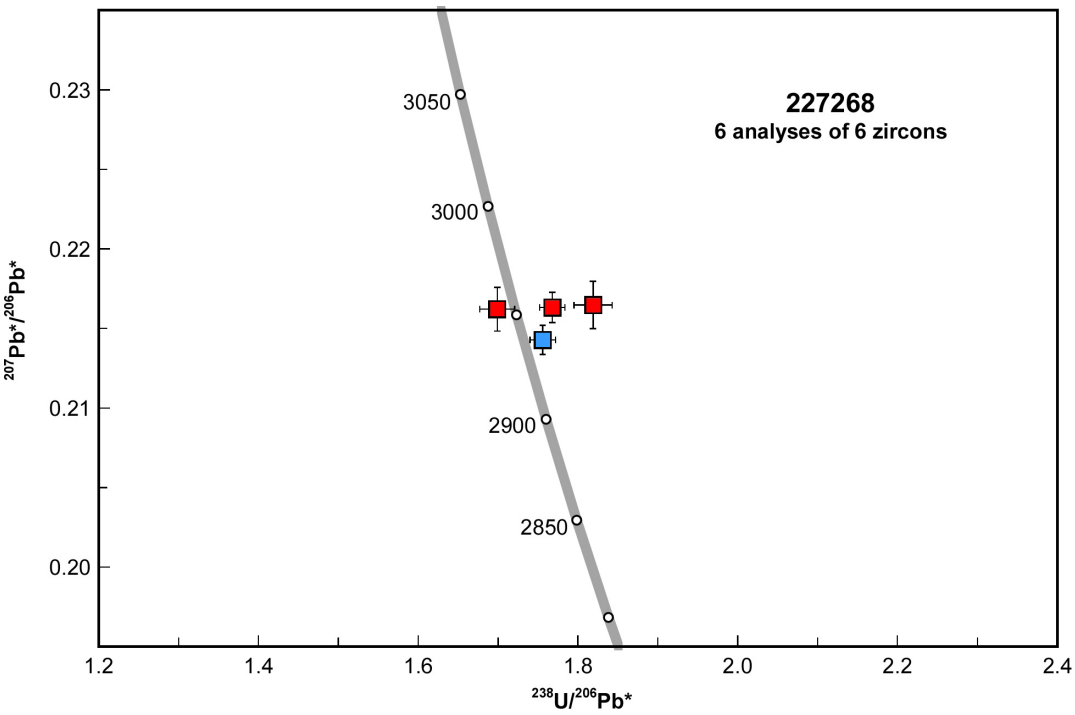

Figure 3. U–Pb analytical data for zircons from sample 227268: altered volcaniclastic rock, Woolgah Hill. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons). Analyses in Groups D and Z are not shown

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). One analysis is >5% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and is considered not to be geologically significant. The remaining five analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2938 ± 7 Ma (1σ).

Group S comprises three analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2955–2953 Ma.

Group Z comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 1665 ± 19 Ma (1σ).

It is possible that all of the analyses in Groups Y and S are of unmodified detrital zircons, in which case the date of 2938 ± 7 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2948 ± 8 Ma (MSWD = 1.1) for the four analyses in Groups Y and S. This date is interpreted as the age of the dominant zircon-crystallizing rock in the detrital source region.

The date of 1665 ± 19 Ma (1σ) for the single analysis in Group Z is interpreted as the age of a zircon contaminant.

References

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Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and Ivanic, TJ 2021, 227268: altered volcaniclastic rock, Woolgah Hill; Geochronology Record 1825: Geological Survey of Western Australia, 4p.

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Data released: 26 November 2021